TikTok Claims Classification Project

Statistical Hypothesis Testing — Executive Summary

■ ISSUE / PROBLEM

TikTok's leadership team requested a statistical analysis to determine whether there is a meaningful difference in video view counts between verified and not verified accounts. This finding will help inform the claims classification model and provide leadership with data-driven insights about account verification and content performance.

■ RESPONSE

A two-sample Welch's t-test was conducted to compare the mean video view counts between verified and not verified accounts. Welch's t-test was selected because it does not assume equal variances between groups — appropriate given the large difference in standard deviations between the two groups.

Hypotheses:

• H■: There is no difference in mean video view count between verified and not verified accounts

• H■: There IS a statistically significant difference in mean video view count between the two groups

• Significance level: $\alpha = 0.05$ (two-tailed test)

■ FINDINGS

Descriptive Statistics

Metric	Verified Accounts	Not Verified Accounts
Sample size (n)	1,155	18,027
Mean view count	88,809	264,332
Median view count	6,134	46,415
Std deviation	221,179	323,334

Hypothesis Test Results

Test Statistic	Value	Decision
t-statistic	-25.2939	
p-value	~0.000000	
Significance level (α)	0.05	
Result		**REJECT H■**

■ IMPACT

The p-value (~0.000) is far below the significance level of 0.05. We reject the null hypothesis — the difference in mean video view counts between verified and not verified accounts is statistically significant and not due to random chance.

Not verified accounts have a mean view count approximately 3x higher than verified accounts. This counterintuitive finding is likely explained by the relationship between verified status and claim status: not verified accounts post more claims, and claim videos generate dramatically higher engagement.

Important: Statistical significance does not imply causation. Verified status should be treated as a correlational signal, not a direct cause of lower viewership.

■ KEY INSIGHTS

The hypothesis test revealed a statistically significant difference in video view counts by verified status. The two key insights from this analysis were:

Verified status affects views

Not verified accounts receive ~3x more views on average than verified accounts (264,332 vs 88,809). This finding is statistically robust ($p \approx 0.000$) and suggests verified status is a meaningful variable for the classification model.

Causation requires further investigation

The view count difference is likely mediated by claim status — not verified accounts post more claims, and claims drive higher engagement. Future regression analysis should model these relationships simultaneously before drawing causal conclusions.